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Accelerated kinetic model for global macro stability studies of high-beta fusion reactors

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Phys. Plasmas 30, 122508 (2023)

<https://doi.org/10.1063/5.0178288>



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Cite as: Phys. Plasmas **30**, 122508 (2023); doi: [10.1063/5.0178288](https://doi.org/10.1063/5.0178288)

Submitted: 26 September 2023 · Accepted: 3 December 2023 ·

Published Online: 19 December 2023



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Note: This paper is part of the Special Topic: Papers from the 2023 Sherwood Fusion Theory Conference.

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ABSTRACT

The field reversed configuration (FRC), such as studied in the C-2W experiment at TAE Technologies, is an attractive candidate for realizing a nuclear fusion reactor. In an FRC, kinetic ion effects play the majority role in macroscopic stability, which allows global stability studies to make use of fluid-kinetic hybrid (also referred to as Ohm's law) models wherein ions are treated kinetically while electrons are treated as a fluid. The development and validation of such a hybrid particle-in-cell algorithm in the Exascale Computing Project code WarpX are reported here. Implementation of this model in the WarpX framework benefits from the numerical efficiency of WarpX as well as its scalability on large HPC systems and portability to different architectures. Performance benchmarks of the new algorithm for large, 3-dimensional, full device simulations from the Perlmutter supercomputer are presented. Results of a series of FRC simulations are discussed in which the impact of two-fluid effects on the tilt-mode growth rate was studied. It was observed that, in agreement with previous Hall-MHD studies, two-fluid effects have a stabilizing impact on the tilt mode.

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I. INTRODUCTION: HIGH-BETA FUSION REACTORS

The push to move fusion research beyond laboratory experiments and toward a commercially viable fusion power plant requires technologists to consider cost and engineering difficulty alongside physics when designing a fusion reactor.¹ As noted in Ref. 2 for magnetically confined fusion concepts, the strength of the applied magnetic field is tightly connected to the cost and engineering difficulty. Furthermore, considering that fusion power scales quadratically with plasma pressure,³ it is clear that one benefits from maximizing the ratio of plasma pressure to magnetic field pressure. This ratio is defined as the plasma β and, as explained in Ref. 2, is a measure of how much plasma pressure can be balanced by a given applied magnetic field. The Sheffield parameter⁴ explicitly includes the volume averaged beta, $\langle\beta\rangle$, in calculating the geometrical and technological requirements for a viable fusion reactor. A fusion reactor operating at high β will necessarily have a relatively low magnetic field (compared to a low- β reactor operating at the same temperature) and therefore relatively low synchrotron radiation, which improves the reactor efficiency and opens the possibility to operate with advanced fuels such as D-³He or p-¹¹B.⁵

The aneutronic p-¹¹B fuel cycle has non-radioactive and superabundant reactants,⁶ making it an appealing prospect both environmentally and economically. These considerations clearly outline the value in exploring high- β concepts.

TAE Technologies Inc. (previously known as "TriAlpha Energy Inc." and hereinafter referred to as "TAE") is pursuing a specific type of a high- β fusion reactor approach based on the field reversed configuration (FRC).^{7,8} A detailed description of TAE's approach and progress can be found in Ref. 9.

A. Macro stability simulation capability

As described in the 2008 FESAC report on alternative (to tokamak) fusion approaches,¹⁰ high- β plasmas present unique challenges that prevent the application of simulation tools commonly used in fusion research to such plasmas. Most crucially, the lack of a strong guiding magnetic field means that flux-coordinates and other gyrokinetic assumptions are invalid. Furthermore, the highly kinetic nature of FRC plasmas necessitates the use of models that retain kinetic effects. In part due to this, there is a lack of community codes applicable for

the study of high- β fusion reactors. For this reason, TAE focused on the development of a proprietary hybrid code called FPIC,¹¹ which over the years has been profitably utilized to investigate different FRC features and phenomena. A non exhaustive list of FRC topics that have been addressed through FPIC simulations includes the $n=1$ toroidal mode,¹² the onset of tilt instability,¹³ the role of fast ion effects on global stability,¹⁴ and the evolution of orbit type distributions.¹⁵ Significant support to the development of FPIC for computationally intensive scenarios was provided by the U.S. Department of Energy through the 2018 INCITE program,¹⁶ through access to the Argonne Leadership Computing Facility. Additional notable work related to FRC stability includes FLAME,¹⁷ HYM,¹⁸ HYPERS,¹⁹ and NIMROD.²⁰ While all of these codes have been used extensively to study the global stability of FRC plasmas,^{21–25} they are not easily accessible to the broader community and do not exploit the most recent advances in HPC (such as multi-GPU executability). Here, we present the development of a fluid-hybrid extension to the open-source particle-in-cell (PIC) code, WarpX,^{26,27} which is shown to enable reactor scale, global stability simulations for FRC plasmas, and to allow TAE to move beyond the previous FPIC code, taking full advantage of the present rapidly evolving exascale computing capabilities. In addition to studying the FRC fusion concept, the hybrid-PIC model is also useful for studying spherical tokamaks,^{28,29} magnetic reconnection,^{30–32} and astrophysical plasmas such as in accretion flow onto supermassive black holes,³³ to name just a few.

Originally, WarpX is a fully kinetic PIC code developed under the Exascale Computing Project (ECP)³⁴ for next generation particle accelerator design. It is built on the architecture agnostic AMReX^{35,36} framework, which endows it with the ability to run on NVIDIA, AMD, or Intel GPUs. The computational efficiency demonstrated in WarpX's PIC routines (it was the first PIC code to run at scale on ORNL's Frontier²⁷) and its open-source development model has made it attractive for extension to other kinetic plasma models beyond fully kinetic, explicit, electromagnetic PIC. This is due to the fact that there is a large overlap in the algorithmic logic between different kinetic models. For example, particle pushing, moment deposition, field gathering, and several other routines use the same logic in different particle based kinetic codes. Reusing these routines in different models allows continuing benefits to be reaped from the optimization efforts of the ECP. At present, beyond fully electromagnetic and the kinetic-fluid hybrid model discussed presently, WarpX also includes electrostatic and magnetostatic capabilities.

The paper is organized as follows: In Sec. II, details of the fluid-kinetic hybrid simulation model are provided with a description of its numerical implementation. Section III describes three tests performed with the code to verify implementation accuracy in 2d and 3d Cartesian geometries. In Sec. IV, the model is applied to study the global stability of an FRC plasma, and the impact of two-fluid effects on the stabilization of the tilt-mode is discussed. Parallel scaling performance results of the model are shown in Sec. V for a problem size suitable for reactor scale simulations. Finally, the paper is concluded in Sec. VI with a summary and outlook for future work.

II. MODEL: HYBRID-PIC

Fusion plasma stability can be divided into macro (global) and micro stabilities, where global stability involves large scale (larger than the ion inertial length) and low frequency (less than the ion cyclotron frequency) disturbances.⁸ On these scales, electron kinetic effects do

not play a significant role and the electron population can be treated as a continuous fluid. In the case of non-relativistic plasmas, the displacement current in the Maxwell–Ampère equation can be neglected³⁷ (effectively letting $\epsilon_0 \rightarrow 0$), which removes speed-of-light effects from the model.³⁸ If one further assumes quasi-neutrality (i.e., $n_e \approx n_i \equiv n$) and neglect terms of order m_e/m_i , the electron momentum equation yields Ohm's law as follows:³⁹

$$\left(en + \frac{m_e}{\mu_0 e} \nabla \times \nabla \times\right) \vec{E} = \vec{J}_e \times \vec{B} - \nabla \cdot \vec{P}_e + en\eta \vec{J} + \frac{m_e}{e^2} (\vec{U}_e \cdot \nabla) \vec{J}_e - \frac{m_e}{e} \left(\frac{\partial \vec{J}_{ext}}{\partial t} + \frac{\partial \vec{J}_i}{\partial t} \right), \quad (1)$$

where $\vec{J} = \vec{J}_e + \vec{J}_i + \vec{J}_{ext}$ is the total current density, $\vec{J}_e = -en\vec{U}_e$ is the electron current density, $\vec{J}_i$ is the ion current density, $\vec{J}_{ext}$ is the external (non-plasma) current density, $\vec{P}_e$ is the electron pressure tensor, n is the plasma density, η is the plasma resistivity, and μ_0 , e , and m_e are the usual natural constants, namely, vacuum permeability, electron charge, and electron mass, respectively.

Neglecting the displacement current term in the Maxwell–Ampère equation gives

$$\mu_0 \vec{J} = \nabla \times \vec{B}, \quad (2)$$

which allows the electron current density, $\vec{J}_e$, to be calculated from the combined knowledge of $\vec{B}$, the ion current density, $\vec{J}_i$ (which is obtained from the standard PIC current deposition of the kinetic ion particles' velocities⁴⁰), and $\vec{J}_{ext}$ via

$$\vec{J}_e = \frac{1}{\mu_0} \nabla \times \vec{B} - \vec{J}_i - \vec{J}_{ext}. \quad (3)$$

Second, the electron pressure is taken to be a scalar, rather than a tensor, and is calculated based on the field free particle kinetics (treating the electrons as an ideal gas) as follows:

$$\vec{P}_e \rightarrow P_e = n_0 T_{e0} \left(\frac{n}{n_0} \right)^\gamma, \quad (4)$$

where n_0 and T_{e0} are the reference density and temperature, respectively, and γ is the adiabatic index with value 1 for the isothermal limit and 5/3 for the adiabatic limit. The electron pressure therefore only depends on the local electron density (which equals the ion density due to the assumption of quasi-neutrality) and can easily be calculated from the deposition of ion charge density. Finally, the electron inertia is neglected, i.e., $m_e \rightarrow 0$. These simplifications result in a state equation for the electric field, which, along with Faraday's law for the evolution of the magnetic field, completes the model

$$\vec{E} = \frac{1}{en} \left[\left(\frac{1}{\mu_0} \nabla \times \vec{B} - \vec{J}_i - \vec{J}_{ext} \right) \times \vec{B} - \nabla P_e \right] + \eta \vec{J}, \quad (5a)$$

$$\frac{\partial \vec{B}}{\partial t} = -\nabla \times \vec{E}. \quad (5b)$$

The kinetic ion particles (which determine $\vec{J}_i$ and n) are treated in the standard PIC manner and pushed using Newton's equation of acceleration and the Lorentz force (with the resistivity term, $\eta \vec{J}$, subtracted from the E -field used in calculating the Lorentz force to conserve momentum; see Ref. 41). A standard particle advancing scheme, such as the Boris-push,⁴² can be used to advance the kinetic particles.

A. Implementation in WarpX

The hybrid-PIC model (Ohm's law solver) as described above has been implemented^{43,44} as an available field solver in the PIC code, WarpX.^{26,27} A description of the algorithm is provided here but can also be found in the WarpX documentation.⁴⁵

1. Ohm's law solver–Algorithm

Inclusion of the hybrid-PIC model in WarpX required implementing the capability to calculate Ohm's law [Eq. 5(a)] for the E -field update (instead of using the Maxwell–Ampère equation as in the fully kinetic PIC). The field solve routine is executed after particle pushing in the PIC loop, which means at step n (such that the simulation time is t_n), the field solver has access to the ρ^{n+1} (charge density) and $\vec{J}_i^{n+1/2}$ (ion current density) from charge and current deposition. The ρ^n and $\vec{J}_i^{n-1/2}$ quantities are also stored from the previous step. The $\vec{E}$ and $\vec{B}$ fields are then updated in three distinct steps.

a. First half step. $\vec{E}^n$ is calculated from Eqs. (3) and (5a) using the ρ^n and $\vec{B}^n$ values as well as an interpolated value $\vec{J}_i^n = (\vec{J}_i^{n-1/2} + \vec{J}_i^{n+1/2})/2$. The $\vec{E}_n$ field is then used to push $\vec{B}^n$ forward in time to $\vec{B}^{n+1/2}$ using Eq. 5(b). The B -field update via the Maxwell–Faraday equation uses the same routine as is used in the “standard” electromagnetic WarpX execution loop.

b. Second half step. Next, $\vec{E}^{n+1/2}$ is calculated with the same routine as before but using $\rho^{n+1/2} = (\rho^n + \rho^{n+1})/2$, $\vec{J}_i^{n+1/2}$ and the updated $\vec{B}^{n+1/2}$. Similarly as before, the magnetic field is then pushed forward to $\vec{B}^{n+1}$ using the $\vec{E}^{n+1/2}$ field.

c. Extrapolation step. The main difficulty in hybrid-PIC codes is to calculate the $\vec{E}^{n+1}$ values.⁴¹ A simple extrapolation step is used in WarpX to calculate $\vec{J}_i^{n+1} = (3\vec{J}_i^{n+1/2} - \vec{J}_i^{n-1/2})/2$, which is used along with ρ^{n+1} and $\vec{B}^{n+1}$ to calculate $\vec{E}^{n+1}$. This closes the field solve logic and allows the PIC routine to continue to the next step. This simple extrapolation does limit the time step that can be taken in the code and is an area of improvement under investigation. Alternative methods include the predictor–corrector approach⁴⁶ and the so-called “current advanced method and cyclic leapfrog” approach.⁴⁷

2. Numerical considerations

The highest frequency waves included in the hybrid-PIC model are the so-called whistler waves (shear Alfvén R-waves⁴⁸), which limit the time step that can be taken when evolving the magnetic field^{41,49} with Eq. 5(b). However, the ion motion does not need to be resolved on the same time step since the necessary step size for the ions simply requires the ion cyclotron motion to be resolved. The algorithm execution time therefore benefits from evolving the magnetic field with a smaller step than is used for particle pushing. This sub-stepping strategy⁴⁷ is implemented in WarpX, wherein both the “first half step” and the “second half step” logics are repeated N times, with the magnetic field advanced $\Delta t/(2N)$ forward in time at every iteration. The value of N is set at runtime based on the problem at hand. The fields are

defined on a standard Yee mesh, but Ohm's law is solved with field values interpolated to a nodal mesh to ensure energy conservation. The hybrid-PIC algorithm currently supports 1D, 2D, and 3D Cartesian geometries and inherits the AMReX embedded boundary functionality already available in WarpX as demonstrated in Sec. III C.

III. BENCHMARKS AND VERIFICATION

A series of tests were performed to validate the hybrid-PIC algorithm implementation in which simulation results were compared to known theoretical results. In this section, several such tests are described with a focus given to both electrostatic and electromagnetic cases. Note that the ion Bernstein modes and ion Landau damping tests follow closely the work from Muñoz *et al.*⁵⁰ The input and analysis scripts for the benchmarks shown here are available as examples on the WarpX Github repository along with more benchmarks as described in the WarpX documentation,⁵¹ including a magnetic reconnection benchmark based on work from Ref. 31.

A. Ion Bernstein modes

Ion Bernstein waves are perpendicularly propagating electrostatic waves. The dispersion relation for pure ion Bernstein waves satisfies the relation

$$1 = 2\omega_{pi}^2 \frac{e^{-\lambda_i}}{\lambda_i} \sum_{n=1}^{\infty} \frac{n^2 I_n(\lambda_i)}{\omega^2 - n^2 \Omega_{ci}^2}, \quad (6)$$

as described in Ref. 52, but with the electron contribution term neglected since, in the current case, $m_e = 0$ is assumed. In this equation, $\omega_{pi} = Ze\sqrt{n}/(\epsilon_0 m_i)$ is the ion plasma frequency, $\Omega_{ci} = ZeB/m_i$, the ion cyclotron frequency, I_n the n 'th order modified Bessel function of the first kind and $\lambda_i = k_{\perp}^2 \rho_i^2$, where $\rho_i = v_{ti}/\Omega_{ci}$ is the ion Larmor radius and $v_{ti} = \sqrt{k_B T_i/m_i}$ is the ion thermal speed. In the above, Z is the ion charge state.

Numerically observing the excitation of these modes is a good test of the accuracy of handling cyclotron resonance effects in the WarpX hybrid-PIC implementation. Generally, PIC noise is sufficient to excite available resonances in a plasma; therefore, a simulation was simply initialized with a uniform plasma and an initial magnetic field (along the x -axis) and evolved for several ion cyclotron periods. The numerical parameters used in the simulation are given in Table I, while the physical parameters are given in Table II. The electric-field values on the simulation grid were output 20 times per cyclotron period. After the simulation reached the set total number of steps, the E_z field values were Fourier transformed in space and time. In this case, the E_z field corresponds to the longitudinal (electrostatic) component of the Bernstein waves. The resulting power spectrum (normalized) is shown in Fig. 1, along with the three lowest harmonics of the fundamental ion Bernstein mode from Eq. (6). The simulation results match the theoretical dispersion well. The dispersion for the lower hybrid mode is also shown, which, as described in Ref. 53, couples well to ion motion perpendicular to an applied magnetic field. At long wavelengths, the dispersion relation for the lower hybrid mode is given by Ref. 54

$$\omega^2 = (v_A^2 + v_s^2)k^2, \quad (7)$$

where $v_A = c\Omega_{ci}/\omega_{pi}$ is the Alfvén speed, and $v_s = \sqrt{\gamma Z}v_{ti}$ is the ion sound speed with γ being the adiabatic index. At short wavelengths, the frequency asymptotically saturates to the lower hybrid frequency ω_{LH} which is given by $\omega_{LH}^2 = \Omega_{ci}^2 + \omega_{pi}^2$ for $m_e = 0$.

TABLE I. Numerical parameters used for the benchmark and verification tests described in Sec. III. The number of cells in each direction, i , is indicated by n_i . In cases where a value is omitted, it is because that direction was not included in the simulation (i.e., 2d simulations). The cell size, dz , is given in units of the ion inertial length, l_i . The simulation time step, Δt , is given in units of the ion cyclotron frequency, Ω_{ci} . The total number of simulation steps is also indicated for each test as well as the initial number of simulation particles per cell used (PPC). Finally, the number of substeps used to update $\vec{B}$ [from Eq. (5)] is given.

	Grid specification					# steps	PPC	Ohm's law solver # substeps
	n_x	n_y	n_z	dz/l_i^a	$\Delta t\Omega_{ci}$			
Ion Bernstein modes (Sec. III A)	8	...	1920	1/10	5×10^{-3}	60 000	256	750
Ion Landau damping (Sec. III B)	4	...	256	1/6	1×10^{-3}	40 000	4096	100
Cylindrical Alfvén modes (Sec. III C)	256	256	512	1/2	2×10^{-2}	50 000	1000	250

^aThe same cell sizes were used in all directions.

Interestingly, it was thought that the lower hybrid instability is responsible for the anomalous particle transport in FRCs, but this was experimentally found to be incorrect⁵⁵ (see also Steinhauer,⁸ Sec C)).

B. Ion Landau damping

Numerical demonstration of the damping of electrostatic plasma waves through particle-wave interactions, the so-called Landau damping,⁵⁶ is commonly used to benchmark plasma codes.^{50,57} This test is particularly useful for hybrid-PIC codes since it isolates the treatment of the electron pressure term as there is no equilibrium magnetic field in this problem. As described in Refs. 57 and 50, the dispersion relation for ion-acoustic waves that undergo ion Landau damping (within the approximations listed in Sec. II) is given by

$$Z'(\zeta_i) = -2 \frac{T_i}{T_e}, \quad (8)$$

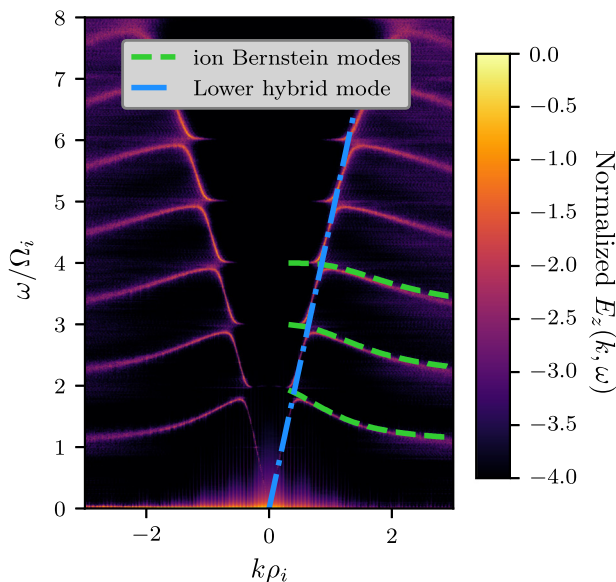


FIG. 1. Normalized power spectrum of an E -field component perpendicular to the applied B -field, showing the harmonics of the fundamental ion-Bernstein mode. The first three harmonics of the theoretical ion Bernstein dispersion from Eq. (6) are also shown as well as the lower hybrid mode dispersion from Eq. (7).

where

$$\zeta_i = \frac{\omega}{k} \sqrt{\frac{m_i}{2k_B T_i}},$$

and Z' is the plasma zeta function⁵⁸ derivative (where the plasma zeta function is the Hilbert transform of a Gaussian). In general, ω calculated from Eq. (8) is complex with its imaginary part, $\gamma = \Im(\omega)$, giving the damping (if $\gamma < 0$) or anti-damping (if $\gamma > 0$) rate. Following the same approach as in Ref. 50, a simulation of a thermal ion population was seeded with a density perturbation that matches a plasma wave, i.e., $\delta n_z = \varepsilon \cos(2\pi m/L_z)$, where L_z is the domain length. An $m=4$ perturbation was used in this case with $\varepsilon = 0.03$, along with the physical simulation parameters given in Table II. Snapshots of the electric field along the direction of the seeded wave were Fourier decomposed to investigate the amplitude of the mode as a function of time. The results of this analysis are shown in Fig. 2 for several cases where the electron temperature used in the Ohm's law solver was varied. The theoretical Landau damping is shown as dashed lines in the figure, confirming that the simulation results show the expected mode evolution in all cases considered.

C. Cylindrical Alfvén modes

Alfvén waves in an infinitely long cylindrical plasma, bounded by a perfectly conducting wall, is discussed in-depth in Stix,⁵⁹ Chap. 6, Sec. II. This problem has also been used by several other authors^{11,60,61} to test implementations of plasma kinetic codes based on Ohm's law. Verifying that the WarpX hybrid-PIC implementation reproduces the correct dispersion for cylindrical waves is particularly important since the code is intended for studying cylindrical reactor geometries, such as the C-2W experiment.⁹ This geometrical constraint is included in a 3d Cartesian geometry via the embedded boundary functionality in WarpX (via AMReX⁶²) to create a metallic cylindrical boundary. Furthermore, in this test, the Hall-term plays a central role and is therefore a good check for implementation accuracy of the curl and cross-product logic, including the handling of these operators at the embedded boundary. All the tests in this section used linear splines when depositing particle quantities onto the simulation grid or interpolating field values to particle coordinates due to the findings reported in Ref. 63 where significant numerical dispersion is seen with hybrid-PIC codes when the ion skin-depth is not well resolved.

As described in Ref. 59, normal modes in a metallic cylinder have solutions of the form,

TABLE II. Physical simulation parameters used in generating the ion Bernstein mode results shown in Fig. 1, the ion Landau damping results shown in Fig. 2, and the cylindrical normal mode results shown in Fig. 3. The ion mass, temperature, and density are given, as well as the background magnetic field and Ohm's law solver parameters, namely, the electron temperature and plasma resistivity.

	m_i/m_e	T_i (eV)	n (m^{-3})	B_0 (T)	Ohm's law	
					T_e/T_i	η ($\Omega \cdot m$)
Bernstein modes	400	10.22	1.5×10^{21}	0.25	1	10^{-5}
Landau damping	100	100	9.6×10^{20}	...	...	10^{-7}
Cylinder modes	1836	20	5.2×10^{20}	1.0	0	10^{-3}

^aVaried as part of the test.

$$\begin{aligned}
 E_r(r, z, t) &= |E_r| \cos(k_z z - \omega t) J_1(\nu_m r), \\
 E_\theta(r, z, t) &= -|E_\theta| \sin(k_z z - \omega t) J_1(\nu_m r), \\
 E_z(r, z, t) &= 0, \\
 B_r(r, z, t) &= -\frac{k_z}{\omega} E_\theta(r, z, t), \\
 B_\theta(r, z, t) &= \frac{k_z}{\omega} E_r(r, z, t), \\
 B_z(r, z, t) &= B_0 + \frac{\nu_m}{\omega} |E_\theta| \cos(k_z z - \omega t) J_0(\nu_m r),
 \end{aligned} \quad (9)$$

with ion current,

$$\begin{aligned}
 J_{i,r}(r, z, t) &= \frac{1}{\mu_0 l_i^2} \frac{\omega}{\Omega_{ci}^2 - \omega^2} \left(\frac{\Omega_{ci}}{\omega} - \frac{|E_r|}{|E_\theta|} \right) E_\theta(r, z, t), \\
 J_{i,\theta}(r, z, t) &= \frac{1}{\mu_0 l_i^2} \frac{\omega}{\Omega_{ci}^2 - \omega^2} \left(\frac{|E_\theta|}{|E_r|} - \frac{\Omega_{ci}}{\omega} \right) E_r(r, z, t), \\
 J_{i,z}(r, z, t) &= 0,
 \end{aligned} \quad (10)$$

where J_n is the n 'th order Bessel function of the first kind, l_i is the ion inertial length, Ω_{ci} is the ion cyclotron frequency and the wave-vectors, and k_z and ν_m are related to ω through

$$\omega^2 = v_A^2 (R^2 \pm \sqrt{R^4 - P^4}), \quad (11)$$

with

$$\begin{aligned}
 R^2 &= \frac{1}{2} [\nu_m (1 + \kappa^2) + k_z^2 (\kappa^2 + 2)], \\
 P^4 &= k_z^2 (\nu_m^2 + \kappa_z^2),
 \end{aligned}$$

where v_A is the Alfvén speed and $\kappa = k_z l_i$. The plus in Eq. (11) corresponds to the so-called fast wave, whereas the minus gives the slow wave. It should be noted that the metallic boundary condition at $r = r_w$ (with r_w the wall radius) requires that $J_1(\nu_m r_w) = 0$, which determines the supported radial wave-vectors, ν_m , based on the roots of J_1 . Axially supported wave-vectors are determined by the cylinder length, L_z , as $k_z = 2\pi n/L_z$. Finally, the relative amplitudes of the E_r and E_θ components are given by

$$\frac{|E_r|}{|E_\theta|} = \frac{\omega^3 / \Omega_{ci}^3}{(\omega^2 / \Omega_{ci}^2)(1 + \kappa^2) - \kappa^2}. \quad (12)$$

1. Dispersion from a thermal ion population

In a similar approach as in Sec. III A, in order to numerically reproduce the dispersion relation from Eq. (11), a thermal ion population was initialized in a simulation with an embedded boundary setup so as to form a cylindrical wall around the plasma (oriented along the z -axis). Periodic boundary conditions were used in the z -direction to mimic an infinite cylinder, and an initial magnetic field was applied along the cylinder axis. The numerical parameters used for the simulation are given in Table I, and the physical parameters are given in Table II. The gridded E_θ values were Fourier transformed in space⁶⁴ and time, and the resulting power spectrum is shown in Fig. 3 (for different radial mode numbers). The theoretical dispersion of cylindrical Alfvén waves is overlaid on the same plot, showing good agreement between the known dispersion and the peaks of the power spectrum from simulation for small k_z values where finite grid effects are negligible.

2. Stability of seeded solution

After confirming that the simulation reproduces the expected normal modes in a cylinder, the evolution of single modes was studied to confirm that they remain stable for many periods. Reference 60 notes that with resistivity included in Ohm's law, the mode will be damped with rate

$$\gamma = \frac{\eta \omega^2}{2\mu_0 v_A^2}, \quad (13)$$

where η is the resistivity value, ω is the mode frequency, and v_A is the Alfvén speed. In this test, a simulation was initialized with the field and ion current profiles given in Eqs. (9) and (10) for specific radial mode, m , and axial mode, n , values. A 1% perturbation was used, i.e., $|E_\theta| = 0.01 B_0 / (\omega \nu_m)$, for the mode amplitude. The simulation parameters were mostly kept the same as in Sec. III C 1 except for the following modifications: the ion temperature, T_i , was set to zero (i.e., no velocity spread); the plasma resistivity was changed to $\eta = \ln(5) \mu_0 v_A^2 / (25\omega)$, in order to more clearly demonstrate the damping of the mode; the cell spacing was changed to $dz/l_i = 1$, the

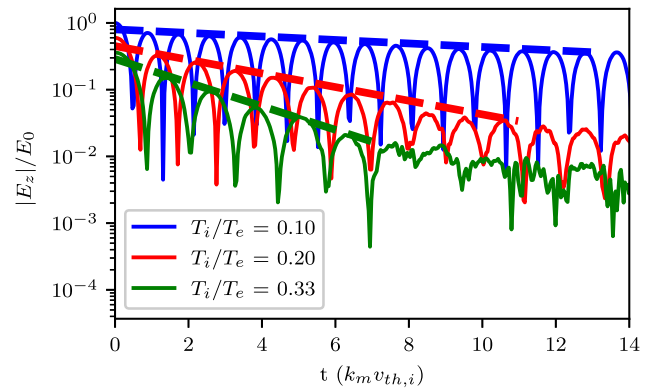


FIG. 2. Amplitude (normalized) of the $m=4$ Fourier mode as a function of time for simulations with different ratios of ion to electron temperature, showing the mode damping over time. The theoretical Landau damping for each simulation is also shown as dashed lines.

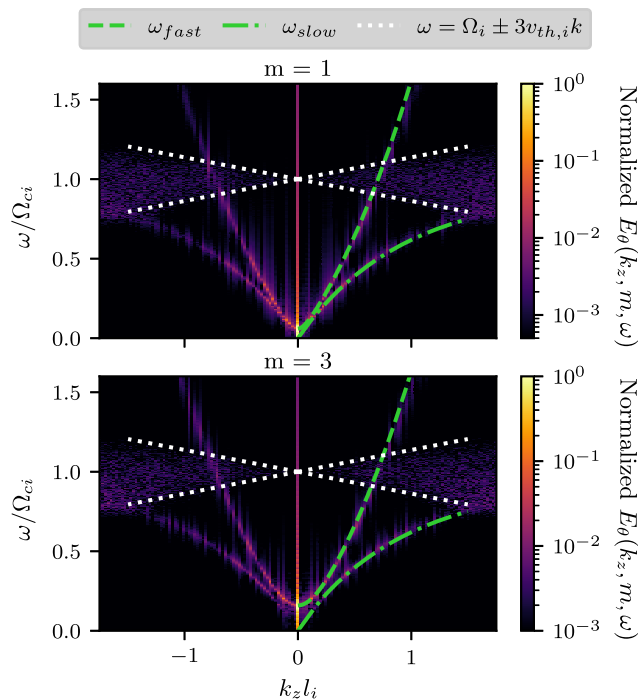


FIG. 3. Normalized power spectrum of the E_θ -field component, showing the dispersion of cylindrical Alfvén waves. Slices through the momentum space at $m=1$ (top) and $m=3$ (bottom) are shown. The theoretical fast and slow dispersions [Eq. (11)] are also plotted as dashed lines, showing that the simulation results match the expectation for both branches. The triangular regions delineated in white are where thermal ions undergo resonance with doppler shifted fields that match their cyclotron frequency.

number of cells reduced to $72 \times 72 \times 144$, and the plasma density was set to $2.6 \times 10^{20} \text{ m}^{-3}$ to broaden the range of parameters used in validating the code. The evolution of B_r at a set point in the simulation domain was tracked as a function of time, as shown in Fig. 4 for two different simulations (seeded with different initial modes). The data were Fourier transformed to measure the mode propagation frequency, which gave $\delta\omega = 0.02$ for the $m=1, n=1$ case and $\delta\omega = 0.04$ for the $m=2, n=3$ case, with $\delta\omega = |1 - \omega_{sim}/\omega|$, where ω_{sim} is the measured frequency and ω is the theoretical frequency from Eq. (11). The evolution of the mode amplitude was also compared with the expected damping rate [Eq. (13)] as shown in Fig. 4 to validate the treatment of the resistivity term in the WarpX Ohm's law solver implementation.

IV. FRC SIMULATIONS

Tilting in FRCs occurs due to the energy minimization from aligning magnetic moments between the internal, “reversed” field and the externally applied field. However, since the plasma in an FRC does not act like a rigid, current-carrying ring, the dynamics of the system are more complicated than that conceptual picture. A recent review of both experimental and simulation results related to the tilt mode can be found in Ref. 8. Stability in FRCs is particularly interesting since MHD predicts them to be grossly unstable,⁶⁵ with the tilt mode causing plasma disruption over a few Alfvén times (defined as the ratio of

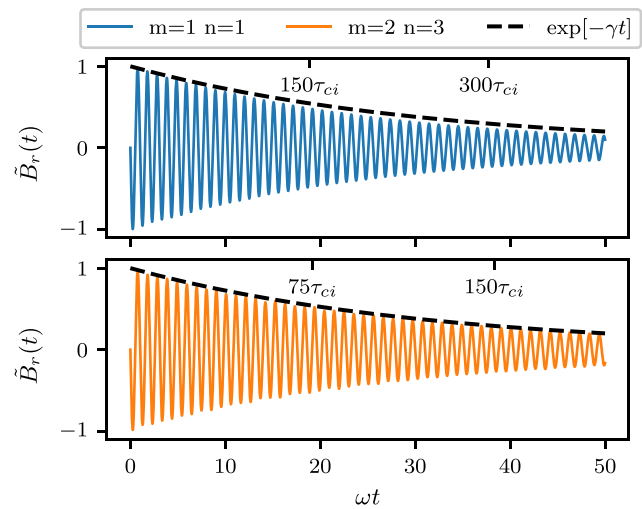


FIG. 4. Evolution of a seeded normal mode in a cylinder [initialized from Eqs. (9) and (10)] with plasma resistivity included, showing cases with $m=1; n=1$ (top) and $m=2; n=3$ (bottom). The B_r -field at $r^* = \max[J_1(\nu_m r^*)]$ and $z=0$ normalized by $(k_z/\omega)|E_\theta J_1(\nu_m r^*)|$ is plotted. Fourier analysis indicates that the mode propagates with the expected frequency [from Eq. (11)] within 4% error and the mode damping matches the expected rate [from Eq. (13)] as indicated by the dashed black line in the plot.

the flux conserving shell radius to the characteristic Alfvén velocity⁶⁶), but experimentally elongated FRCs are observed to survive much longer than this. Reference 8 reports that 2/3 of the data in the FRC data compendium⁶⁷ show particle confinement times between 21 and 63 times the MHD tilt mode growth period, which is typically a few μs ($1/\gamma_{\text{MHD}} \approx L_s/(2Cv_A)$, where L_s is the separatrix length, v_A is the characteristic Alfvén velocity, and C is a constant that depends on the equilibrium profile). The C-2W experiment has recorded FRC lifetimes longer than 25 ms.⁹ This contradiction between the MHD theory and experiment is largely resolved through the inclusion of finite-Larmor radius (kinetic ion) and two-fluid (separation of electron and ion responses) effects. Several authors have reported large reductions in the tilt growth-rate observed in simulations that capture finite-Larmor radius (FLR) and two-fluid effects vs the MHD growth rate for elongated FRCs with a high kinetic parameter ($S^* = R_s/l_i$, where R_s is the separatrix radius).^{18,22,68} For this reason, the hybrid-PIC method is well suited to study the global stability of FRCs and the tilt mode, in particular.

Belova *et al.*⁶⁸ studied the impact of the Hall effect in stabilization of the tilt mode using a Hall-MHD model while neglecting the electron pressure (*i.e.*, $T_e = 0$). Similar earlier studies were also performed using a variational approach with trial functions.^{69,70} These studies found a reduction in the tilt growth rate due to the Hall effect for FRC equilibria with low S^* value but did not find complete stabilization, which led to the conclusion that finite-Larmor-radius effects are primarily responsible for the non-linear saturation of the tilt mode observed in hybrid-PIC simulations⁷¹ with $S^*/E \sim 1$. In the present study, we extend on these earlier works by studying the combined two-fluid effects, Hall and finite electron pressure, using the newly developed full- f WarpX hybrid-PIC module. A similar approach is used as in Ref. 32 where a numerical parameter is inserted to group two-fluid effects

in Ohm's law as indicated below. In this study, the B -field update used a modified version of the Maxwell–Faraday equation, given by

$$\frac{\partial \vec{B}}{\partial t} = \nabla \times \left(\frac{1}{en} \vec{j}_i \times \vec{B} - \frac{\alpha}{en} \left[\frac{1}{\mu_0} \nabla \times \vec{B} \times \vec{B} - \nabla P_e \right] \right), \quad (14)$$

where α is the numerical parameter used to modify the strength of the two-fluid terms. This modification is only applied to the B -field update since including it in the Lorentz force applied to the ions would violate the momentum equation.³⁰

A. Initial FRC equilibrium

An equilibrium FRC profile generated from the model described in Ref. 72 was used for the initial conditions in this study. A fixed initial equilibrium was used for all the FRC simulations described below, which allows the impact of the value of α (i.e., the strength of two-fluid effects) to be fully separated from equilibrium profile effects. An equilibrium with an elliptic separatrix was used with $L_s = 2.6$ m and $R_s = 0.5$ m, while the equilibrium density profile was such that $S^* = 6.5$ (giving $S^*/E = 2.5$). The external magnetic field strength was 0.4 T close to the cylinder wall at the mid-plane, with magnetic mirrors on both ends of the cylinder. The equilibrium current was fully carried by the electrons, i.e., there was no ion rotation in the equilibrium.

B. FRC simulation parameters

A $72 \times 72 \times 256$ grid with a 2.5 cm cell resolution was used for all the FRC simulations in this section. Periodic boundary conditions were applied in the z -direction for both fields and particles, while a conducting embedded boundary (absorbing for particles) was used to mimic a cylindrical vessel, similar to the setup used in Sec. III C. The time step was set to 1 ns, and the simulation was initialized with 1.4×10^9 macroparticles (~ 1050 particles per cell) distributed

according to the density and temperature profiles from the calculated equilibrium. Particle velocities were sampled from a Maxwellian distribution with the local temperature based on the equilibrium profile. The Esirkepov current deposition algorithm was used with quadratic splines for the deposition and interpolation of grid quantities to reduce noise. These simulations were run with a single precision floating point arithmetic to reduce the required data transfer between GPUs.⁷³ All simulations were performed on the Perlmutter supercomputer using 1 GPU node, i.e., 4 A100 NVIDIA GPUs.

C. Hybrid-PIC simulations of tilt mode evolution

The evolution of the hybrid-PIC simulation with $\alpha = 1$ is shown in Fig. 5 as snapshots at different times. In agreement with literature reports, in the initial phase of the simulation, ion spin-up ($m = 0$ azimuthal mode) is observed in the ion-diamagnetic direction²² followed by a slow development of the tilt mode ($m = 1$). By $t = 60t_A$, the tilt amplitude peaks (see also Fig. 6), with both the plasma density and axial current profiles showing a clear $m = 1$ signature. The FRC is fully disrupted by $t = 85t_A$, which is much later than the MHD predicted lifetime of $1/\gamma_{MHD} \approx 1.6t_A$.

In order to study the impact of two-fluid effects, a series of simulations were performed with different α values, including $\alpha = 0$, in which two-fluid effects are turned off as well as $\alpha > 1$ as a computational experiment of what the effect of an enhancement of these terms would produce. Higher α values resulted in a slower disruption of the FRC, suggesting that two-fluid effects do impact the tilt growth rate. The amplitude of the tilt mode was calculated as a function of time for each simulation following the approach described in Ref. 74 (see also Ref. 12), wherein the symmetric axial ion-velocity field (over the $z = 0$ mid-plane) is calculated

$$v_z^+(\theta, z) = \int_0^{r_{\max}} (v_z(\theta, z) + v_z(\theta, -z)) dr, \quad (15)$$

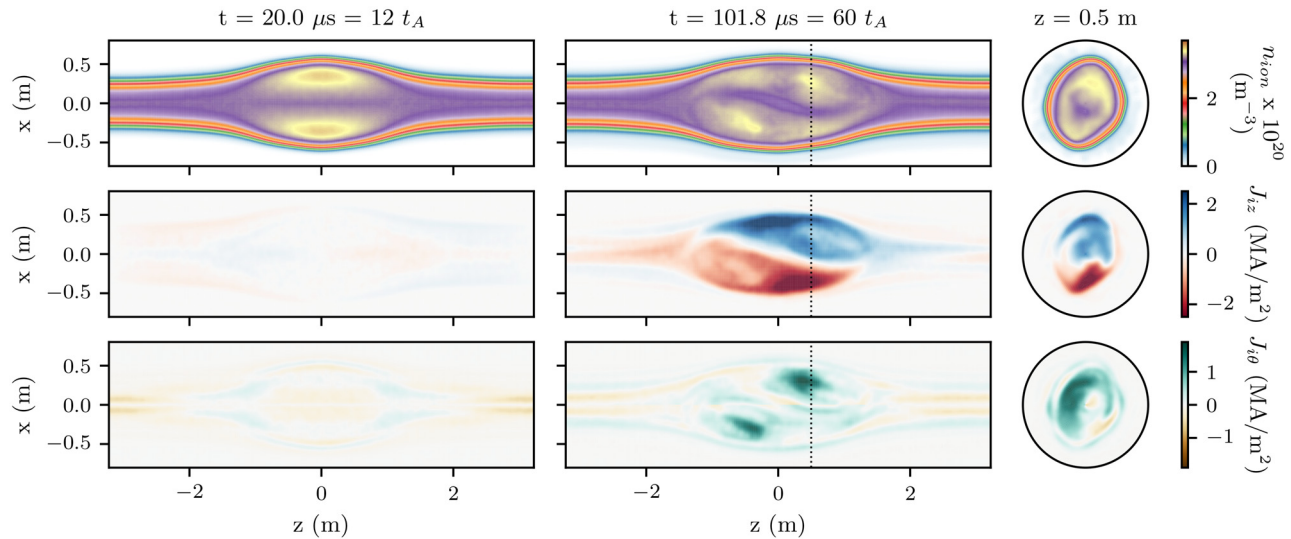


FIG. 5. Evolution of the hybrid-PIC simulation with $\alpha = 1$. Slices at $y = 0$ are shown for two times, early in the simulation before tilt develops as well as later in the simulation when the tilt mode saturates. Polar plots at $z = 0.5$ m are also shown for the later time, in which strong $m = 1$ mode activity is visible in both the ion density and axial ion current density.

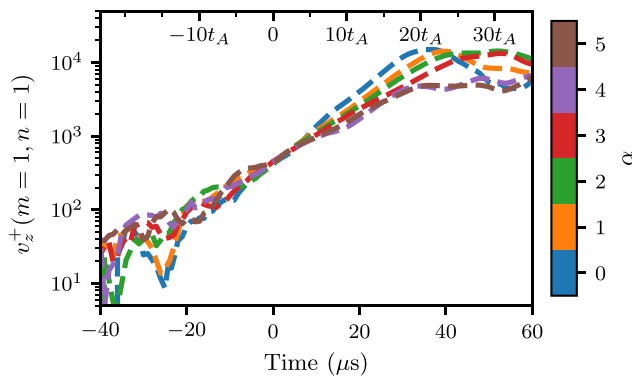


FIG. 6. Amplitude of the $m=1$ azimuthal, $n=1$ axial mode of v_z^+ field [see Eq. (15)] as a function of time for a series of simulations starting from the same equilibrium but with different α [strength of two-fluid effect, see Eq. (14)] values, showing that the linear growth rate of the mode amplitude is lower for higher α values.

and Fourier decomposed into axial and azimuthal modes. The $m=1$, $n=1$ mode corresponds primarily to the tilt mode. The $v_z^+(m=1, n=1)$ amplitudes for the series of simulations with different α values are shown in Fig. 6. The results were shifted in time from the actual simulation time to overlap at the start of the linear growth phase since these simulations were not seeded with a mode but instead relied on PIC noise to initialize tilt. Visual inspection of the relative growth rates of the mode amplitudes indicates that higher α values correspond to a lower linear growth rate. This is confirmed quantitatively through a fit of the slopes of the mode amplitude curves as shown in Fig. 7. The growth rates were fitted between the shifted $t=0$ point and the time at which the mode amplitude increased to $20\times$ its starting value ($15\times$ for $\alpha \in [4, 5]$ since the mode amplitude saturated at less than $20\times$ its $t=0$ value). Error bars for the fitted growth rates were determined by changing the end point for the range over which the slope is fit and finding the maximum and minimum values for the growth rates. This range of end points ran from $5\times$ to $20\times$ of the amplitude value at $t=0$. The results of this series of simulations show that the tilt-mode growth rate is reduced by the inclusion of two-fluid

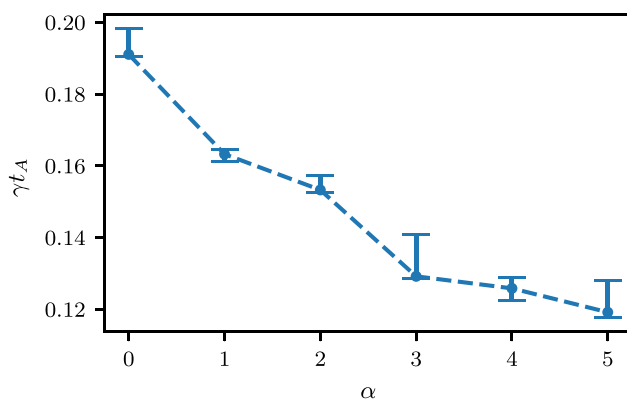


FIG. 7. Fitted linear growth rates of the mode amplitudes from Fig. 6, showing a decrease in the mode growth rate with an increase in α . The error bars were calculated by determining the maximum and minimum fitted growth rates over a series of fitting ranges as described in the main text.

effects. This work extends on the results of Belova *et al.*⁶⁸ in which the effect of inclusion of the Hall-term only (*i.e.*, no electron pressure) was studied with a Hall-MHD model and it was also found that the Hall-term decreases the tilt-mode growth rate.

V. GPU ACCELERATED CODE PERFORMANCE

In anticipation of applying WarpX to study the global stability of fusion grade, high- β plasmas, a parallel performance study of the code was conducted. A kinetic analysis of reactivity with updated p-¹¹B cross-sections^{75,76} found that an ion temperature of 200–400 keV is required for a high Q reactor.⁶ At fusion relevant conditions, the ion density will also be higher than in the problem described in Sec. IV, leading to a decrease in the ion skin-depth, which necessitates an increase in the number of grid cells in each dimension. It is estimated that a grid size of $148 \times 148 \times 592$ will be sufficient to achieve the required spatial resolution to model a future fusion reactor. A problem of this size was run on the NERSC Perlmutter supercomputer using different numbers of GPU nodes (with 4 A100 NVIDIA GPUs per node⁷⁷) and changing the number of particles in the simulation based on the number of GPUs available. The performance tests used particle shape 1, the Esirkepov current deposition algorithm, and single precision particle quantities. The simulations were run for 1000 steps to get the average timing data per step with diagnostics writing the E, B, J , and ρ fields to file every 250 steps. The results of this scaling study are shown in Fig. 8 with a breakdown of the time per step spent in major parts of the algorithm. The highly optimized particle handling in WarpX is evident in the almost perfect weak scaling observed in the “particle push” time. The time spent in advancing the electromagnetic fields decreases with an increase in the number of nodes used since the grid-size remains fixed for all cases. However, it appears that the benefit of more available GPUs on the field solve time saturates at 32 GPUs since the time per step spent on the field solve with 32 GPUs (480 particles-per-cell) is about the same as with 64 GPUs (960 particles-per-cell). It should be noted that the deposition of particle moments (current and charge density) is counted in the “field solve” time as these quantities are used during the calculation of the fields. Diagnostic output uses the ADIOS2 library,⁷⁸ which also benefits from

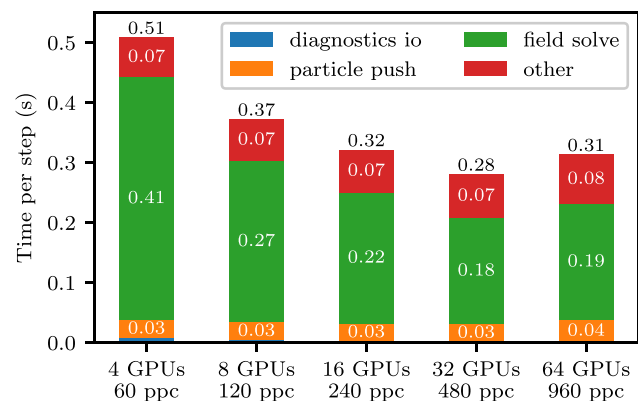


FIG. 8. Parallel scaling of the WarpX hybrid-PIC algorithm on Perlmutter GPU nodes with a fixed grid-size of $148 \times 148 \times 592$, while particle count increases proportionally to the compute resources, *i.e.*, a mixed strong and weak scaling test.

the fixed grid size, resulting in almost negligible time to output data at a high node count.

VI. DISCUSSION AND FUTURE OUTLOOK

In order to best leverage new HPC clusters with increased GPU resources and compute power for FRC simulations, we have extended the simulation capabilities of the open-source ECP code WarpX through the addition of a hybrid PIC algorithm. The implementation has been validated against multiple standard plasma physics problems, such as the propagation of ion Bernstein modes and Landau damping. Additionally, a benchmark of Alfvénic normal modes in a cylindrical plasma verifies the functionality of a cylindrical embedded boundary that is relevant for FRC fusion reactor simulations.

A hybrid PIC model is useful for 3D, reactor-scale global stability studies of FRCs as the model resolves ion kinetic effects that are important in the global stability of FRCs while stepping over the electron kinetics. This allows saving time and computational resources as compared to fully kinetic PIC simulations in which the plasma Debye length must be resolved. Seeing as the hybrid-PIC scheme only needs to resolve the ion inertial length, the spatial resolution can be $l_i/\lambda = c/v_{th}$ (assuming equal electron and ion temperatures) times larger (typically 10^2 – 10^3), which allows at least the same increase in the required time step. Therefore, while a typical fully kinetic (explicit) PIC simulation of an FRC would be limited to a time step of $\sim 10^{-12}$ s, a hybrid-PIC simulation can use a time step of $\sim 10^{-9}$ s and consequently would reach the same simulation evolution time ~ 1000 times faster. Results were presented in which the WarpX hybrid PIC model was used to study changes to the tilt mode growth rate as the strength of the two-fluid terms in Ohm's law was modified. These simulations showed a decrease in the tilt growth rate due to two-fluid effects, which is in agreement with an earlier, similar study conducted with Hall-MHD simulations. Furthermore, this parameter sweep demonstrates how the hybrid model can easily push simulations into new physics regimes, which are of interest to explore.

Exploring new reactor regimes will also be possible due to productivity gains from using ECP and community codes like WarpX. By employing an open-source code for science exploration, TAE benefits from the optimization of performance across multiple CPU and GPU hardware architectures (AMD, Intel, and NVIDIA) continually produced by the WarpX community. The code is also actively used on many different HPC clusters (including Frontier, Fugaku, and Perlmutter), which makes WarpX easily deployable⁷⁹ and simplifies the process of migrating between clusters for different research projects.

Future work on the hybrid solver will focus on improving model accuracy, performance, and capturing additional physics. Hybrid PIC codes with inertialess electron fluids suffer from numerical instability in low density regions due to the dependence of the E-field on the inverse density. Inclusion of a finite electron mass to resolve such issues will be considered (see, for example, Refs. 49 and 50). Additionally, implementing anisotropic electron pressure will allow the model to simulate more plasma regimes specifically of interest for magnetic reconnection studies. An extension of the model to a 2D-3V RZ implementation is also planned in which the field quantities will be calculated on an axisymmetric grid rather than a Cartesian grid, which will allow better resolution of azimuthal modes, as well as other algorithmic improvements to increase performance. These efforts are part of TAE Technologies' goal of establishing a whole-facility modeling

framework, which is an important step toward designing and building a functioning fusion power plant.

ACKNOWLEDGMENTS

This research used resources of the National Energy Research Scientific Computing Center, a DOE Office of Science User Facility supported by the Office of Science of the U.S. Department of Energy under Contract No. DE-AC02-05CH11231 using NERSC award NERSC DDR-ERCAP0025562.

This research used the open-source particle-in-cell code WarpX <https://github.com/ECP-WarpX/WarpX>, primarily funded by the U.S. DOE Exascale Computing Project. Primary WarpX contributors are with LBNL, LLNL, CEA-LIDYL, SLAC, DESY, CERN, and TAE Technologies. We acknowledge all WarpX contributors.

AUTHOR DECLARATIONS

Conflict of Interest

Authors R.E.G., A.V., A.N., B.S.N., T.T., and S.D. are employees of TAE Technologies. Authors FC and DB are paid consultants for TAE Technologies. TAE Technologies is a privately funded company working toward commercialization of a nuclear fusion power plant.

Author Contributions

Roelof Erasmus Groenewald: Conceptualization (equal); Formal analysis (lead); Methodology (equal); Software (lead); Validation (lead); Visualization (equal); Writing – original draft (lead). **Avigdor Veksler:** Data curation (equal); Formal analysis (equal); Validation (equal); Visualization (equal); Writing – review & editing (equal). **Francesco Ceccherini:** Conceptualization (equal); Formal analysis (supporting); Investigation (equal); Methodology (equal); Validation (supporting); Writing – review & editing (equal). **Ales Necas:** Conceptualization (equal); Formal analysis (supporting); Investigation (equal); Methodology (supporting); Writing – review & editing (equal). **Bradley Scott Nicks:** Data curation (supporting); Formal analysis (supporting); Software (supporting); Validation (supporting); Writing – review & editing (equal). **Daniel C. Barnes:** Conceptualization (equal); Methodology (lead); Software (supporting); Writing – review & editing (equal). **Toshiki Tajima:** Funding acquisition (lead); Project administration (equal); Resources (equal); Supervision (supporting); Writing – review & editing (supporting). **Sean Dettrick:** Conceptualization (lead); Investigation (equal); Methodology (equal); Project administration (equal); Resources (equal); Supervision (lead); Writing – review & editing (equal).

DATA AVAILABILITY

The data that support the findings of this study are openly available in WarpX Github repository at <https://github.com/ECP-WarpX/WarpX.git>, Ref. 80. The data that support the findings of this study are available from TAE. Restrictions apply to the availability of these data, which were used under license for this study. Data are available from the authors upon reasonable request and with the permission of TAE.

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